HDOC Day-44

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Algorithm

Main Program

1. Start.
2. Display the menu:
    - Reverse each word in a sentence.
    - Print the initials of a name.
    - Exit.
3. Read the user's choice.
4. Use `switch-case` to call the appropriate function.
5. Repeat the menu until the user selects Exit.
6. Stop.

Operation 1 — Reverse Each Word

1. Read a sentence using `fgets()`.
2. Start scanning the sentence from the first character.
3. Find the beginning and end of each word.
4. Reverse the characters of that word by swapping the first and last characters, then the second and second-last characters, and so on.
5. Do not change the positions of the words.
6. Continue until the end of the sentence.
7. Display the modified sentence.

Example:

```
Input : What is special?
Output: tahW si ?laiceps
```

Note: Reversing characters preserves their original case, so `What` becomes `tahW`.

Operation 2 — Print Initials

1. Read the full name using `fgets()`.
2. Scan the name from left to right.
3. Print the first alphabetic character as an initial.
4. Whenever a new word starts after a space, print its first alphabetic character.
5. Convert the initial to uppercase using `toupper()`.
6. Print a `.` after every initial.

Example:

```
Input : Alexander Grahambell
Output: A.G.
```

Test-Case Table

Test	Choice	Input	Expected Output
1	1	`What is special?`	`tahW si ?laiceps`
2	1	`Hello World`	`olleH dlroW`
3	1	`C Programming`	`C gnimmargorP`
4	1	`Good Morning Everyone`	`dooG gninroM enoyrevE`
5	2	`Alexander Grahambell`	`A.G.`
6	2	`Aaryan Pratap Singh`	`A.P.S.`
7	2	`john smith`	`J.S.`
8	2	`Ratan Naval Tata`	`R.N.T.`
9	3	—	Program exits
10	5	—	Invalid choice message

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

#define MAX 200

void reverseEachWord(void);
void printInitials(void);

int main(void)
{
   int choice;

   do
   {
      printf("\n===== HDOC Day-44 =====\n");
      printf("1. Reverse each word in a sentence\n");
      printf("2. Print the initials of a name\n");
      printf("3. Exit\n");

      printf("Enter your choice: ");
      scanf("%d", &choice);
      getchar();
```

```
        switch (choice)
        {
            case 1:
                reverseEachWord();
                break;

            case 2:
                printInitials();
                break;

            case 3:
                printf("Exiting program...\n");
                break;

            default:
                printf("Invalid choice. Please enter 1, 2, or 3.\n");
        }

    } while (choice != 3);

    return 0;
}


/* Function to reverse each word */
void reverseEachWord(void)
{
    char sentence[MAX];
    char temp;
    int i, start, end;
    int length;

    printf("Enter a sentence: ");
    fgets(sentence, sizeof(sentence), stdin);

    sentence[strcspn(sentence, "\n")] = '\0';

    length = strlen(sentence);
    i = 0;
```

```
    while (i < length)
    {
        /* Skip spaces */
        while (i < length && isspace((unsigned char)sentence[i]))
            i++;

        start = i;

        /* Find end of current word */
        while (i < length &&
               !isspace((unsigned char)sentence[i]))
            i++;

        end = i - 1;

        /* Reverse current word */
        while (start < end)
        {
            temp = sentence[start];
            sentence[start] = sentence[end];
            sentence[end] = temp;

            start++;
            end--;
        }
    }

    printf("Reversed sentence: %s\n", sentence);
}


/* Function to print initials */
void printInitials(void)
{
    char name[MAX];
    int i;
    int newWord = 1;

    printf("Enter the name: ");
    fgets(name, sizeof(name), stdin);
```

```
    printf("Initials: ");

    for (i = 0; name[i] != '\0'; i++)
    {
        if (isspace((unsigned char)name[i]))
        {
            newWord = 1;
        }
        else if (newWord &&
                 isalpha((unsigned char)name[i]))
        {
            printf("%c.",
                   toupper((unsigned char)name[i]));

            newWord = 0;
        }
    }

    printf("\n");
}
```

Execution

